Algebra Readiness Check · Form B

Name _____ Class _____ Date _____

Choose the best answer and circle its letter. Show any work in the space beside the question. This check is not graded; it helps your teacher see how you are thinking, so a careful guess is better than a blank.

1. Which equation is a correct next step for solving $5x + 3 = 2x + 15$?

   A $7x + 3 = 15$
   B $5x + 3 = 17x$
   C $5x = 2x + 18$
   D $3x + 3 = 15$

2. For which values of x is $2(x + 3) = 2x + 6$ true?

   A Only $x = 0$
   B No values, because the x-terms cancel
   C Every real number
   D Only $x = -3$

3. n can be any whole number. Which expression is ALWAYS an even number?

   A $n + 2$
   B e
   C $2n + 1$
   D $2n$

4. Which is larger, $2n$ or $n + 2$?

   A $2n$, because multiplying makes a number bigger
   B They are equal
   C $n + 2$, because adding 2 always gives more
   D It depends on the value of n

5. Is $a + b + c = a + p + c$ always true, never true, or sometimes true?

   A Always true
   B Never true, because b and p are different letters
   C Sometimes true, when $b = p$
   D Never true, because there are no numbers to check

6. Which expression is equivalent to $(3x + 6) / 3$?

A $x + 6$
B $3x + 2$
C $3x$
D $x + 2$

7. Which expression is a factored form of $x^2 - 9$?

A $(x - 3)^2$
B $(x - 3)(x + 3)$
C $(x - 9)(x + 1)$
D $x(x - 9)$

8. Solve $3(x - 2) = x + 4$.

A $x = 3$
B $x = 5$
C $x = 5/2$
D There is no solution, because x is on both sides.

9. The formula $d = rt$ relates distance, rate and time. Which expression gives t?

A $t = d / r$
B $t = dr$
C $t = d - r$
D $t = r / d$

10. Solve $x/3 + 2 = 5$.

A $x = 1$
B $x = 13$
C $x = 9$
D $x = 21$

11. What is the value of 2^{-3}?

A -8
B -6
C $1/8$
D $1/6$

12. Which expression is equivalent to $x/3 + x/4$?

A $2x/7$
B $x/7$
C $7x/12$
D $x^2/12$

13. The table shows values of a linear function: x = 0, 1, 2, 3 gives y = 7, 4, 1, −2. Which equation fits the table?

A $y = 3x + 7$
B $y = 7x - 3$
C $y = -3x + 7$
D $y = -3x$

14. A graph shows Maya's distance from home, d (km), against time, t (hours). From t = 1 to t = 2 the graph is a horizontal segment at d = 3. What was Maya doing during that hour?

A Staying in one place, 3 km from home
B Walking along a flat road
C Moving at a constant speed of 3 km per hour
D Travelling 3 km during that hour

15. A phone plan costs C = 15 + 0.10m dollars for m minutes of calls. What does 0.10 represent?

A The monthly starting fee
B The total cost of m minutes
C The number of minutes used
D The cost of each minute, in dollars

16. What is the domain of f(x) = 1 / (x − 4)?

A All real numbers except 0
B All real numbers greater than 4
C All real numbers except 4
D All real numbers except −4

17. For a function h, h(2) = 7. What does this statement mean?

A When the input is 7, the output is 2.
B When the input is 2, the output is 7.
C h times 2 equals 7, so h = 3.5.
D The rule for h is 'add 5'.

18. A car rental costs $40 plus $25 per day. Which expression gives the total cost, in dollars, for d days?

A 25 + 40d
B 40 + 25d
C 65d
D 40d + 25d

19. Which expression represents 'three less than twice a number n'?

A 3 − 2n
B 2(n − 3)
C 3 < 2n
D 2n − 3

20. The price p of a jacket increases by 20%. Which expression gives the new price?

A 1.2p
B p + 20
C 0.2p
D p / 1.2